

Configurator Tutorial

A complete, example-driven guide to profiles, subprofiles, OLED labels, RGB, macro steps, device synchronization, backups, multiple decks, and repair.

Import

ˆ
> Edit

ˆ
> Compare

ˆ
> Preview

ˆ
> Sync

The one rule to remember

Import Device copies from the MacroPad into the editor. Sync Device copies from the editor to the MacroPad. Use Compare before Sync when the device contains anything important.

Applies to configurator version 1.1 and the matching CircuitPython firmware. Generated from the project tutorial source.

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1. Start here: the configurator mental model

The desktop configurator is an editor and deployment tool. The MacroPad does not need the desktop application after synchronization. The CircuitPython firmware on the device executes keyboard shortcuts, typed text, media controls, mouse actions, delays, OLED labels, and RGB behavior on its own.

The three copies you may encounter

Copy	Where it lives	What it means
Local editor library	Qt application-data folder on the computer	The working copy displayed and autosaved by the GUI.
Device library	The selected MacroPad CIRCUITPY drive	The profiles currently used by that device.
Export or backup	A file chosen by you or the per-device backup folder	A portable library archive, one-profile JSON, or historical snapshot.

Direction matters

Import Device: device -> local editor.

Sync Device: local editor -> device.

Export and backups do not change the connected device until you later choose Sync Device.

Status badge meanings

Status	Meaning	Next action
Autosaving...	An edit is waiting for the 750 ms local autosave timer.	Wait briefly or choose Save Local.
Local saved - device differs	The working copy is safe on the computer but not deployed.	Compare, preview if useful, then Sync Device.
Synced/local saved	The selected device and local normalized libraries match.	No action is required.

Tooltips are built into the application

Hover over any menu action, toolbar command, profile button, key tile, input, palette control, or macro-step field. Every tooltip contains a short purpose and a concrete Example line. Tooltips are the fastest in-context reminder; this PDF provides the complete workflow and the reasons behind each command.

2. Installation and launch

Requirements

- Linux desktop with Python 3.10 or newer.
- PySide6 and pyserial, installed through the project package.
- A normal USB data cable. Charge-only cables cannot mount CIRCUITPY or expose serial.
- For firmware repair, circup must be installed in the same application environment.

Create the environment once

```
python3 -m venv --system-site-packages .venv
.venv/bin/pip install -e '.[test]'
```

Launch options

Method	Command or action	Use when
Project launcher	<code>./run_gui.sh</code>	Running from a terminal in the repository.
File-manager launcher	Double-click <code>Start-MacroPad-Configurator.sh</code>	Starting without an open terminal.
Installed command	<code>macropad-configurator</code>	The package is installed into the active environment.
Exit	File -> Exit or Ctrl+Q	Closing through save and preview-cleanup checks.

Connect the MacroPad

Connect a MacroPad running CircuitPython normally. Its drive should mount as CIRCUITPY and contain `boot_out.txt` with the Adafruit MacroPad RP2040 board ID. A board in UF2 bootloader mode appears as RPI-RP2 and is not treated as a configurable MacroPad.

1. Wait until the CIRCUITPY drive is mounted.
2. Launch the configurator.
3. Choose Refresh if the device was connected after launch.
4. Select the correct device by friendly name, UID, and mount path.

3. Complete interface tour

The window is organized as a top device toolbar, a second editing toolbar, and three working columns. The left column manages parent profiles. The center column manages layout screens and the 12-key pad. The right column edits the selected control's label, RGB, and action sequence.

Region	Primary job	Typical example
Device toolbar	Select a device; refresh, import, preview, sync, save, and choose keyboard layout.	Select Left App Deck, then Preview RGB.
Editing toolbar and menus	Undo, copy/paste, swap, compare, backups, import/export, health, repair, naming, and exit.	Compare the local library before syncing.
Left: Profile screens	Create, select, delete, import, export, template, and reorder parent profiles.	Move Firefox next to VS Code in encoder-turn order.
Center: Active profile	Name and reorder subprofiles, set icon and brightness, preview OLED, and select keys.	Create Browse, Tabs, Tools, and In App layouts.
Right: Control editor	Name a key, set its six-character OLED label, configure RGB, and build macro steps.	Create a green Copy key with CONTROL+C.

Physical key numbering

Key 1 row 1	Key 2 row 1	Key 3 row 1
Key 4 row 2	Key 5 row 2	Key 6 row 2
Key 7 row 3	Key 8 row 3	Key 9 row 3
Key 10 row 4	Key 11 row 4	Key 12 row 4
Encoder press - below the 12 keys		

Keys are numbered left-to-right and top-to-bottom. The first row is Keys 1-3, the second is 4-6, the third is 7-9, and the fourth is 10-12. This matters when you choose Current row in the palette scope or read profile documentation.

Parent profiles versus layout screens

A parent profile is selected by turning the encoder. A layout screen, also called a subprofile, is selected by pressing the encoder while that parent is active. Each layout has its own 12 keys, icon, brightness, colors, labels, and steps.

Turn encoder ➤ Choose parent ➤ Press encoder ➤ Choose layout ➤ Press key

Encoder assignment rule

A parent with only one layout may use a custom encoder-press macro. As soon as additional layouts exist, encoder press is reserved for cycling layouts and its action editor is disabled.

4. First-device workflow: preserve, edit, verify, deploy

Use this workflow when the MacroPad already contains anything worth keeping. It makes direction explicit and gives you two review points before a write.

Import Device ➤ Edit ➤ Compare ➤ Preview RGB ➤ Sync Device

Step 1 - Import Device

1. Select the intended MacroPad in the Device list.
2. Choose Import Device.
3. If prompted about local edits, stop unless you intend to replace them.
4. Confirm the profiles visible in the left panel match the device.

Import replaces the local working library. It does not change the MacroPad. If the local library also matters, export it before importing.

Step 2 - Make a small, identifiable edit

For a first test, choose a harmless key. Rename it, set a short OLED label, and change its idle color. A visible but safe change is easier to verify than a complex launcher or shutdown macro.

Step 3 - Compare

Compare reports keyboard-layout changes, profile membership or order, names and icons, brightness, changed controls, and subprofile changes. Read the summary. Unexpected removals usually mean you imported or selected the wrong library.

Step 4 - Preview RGB

Preview RGB sends temporary colors for the selected layout. It does not deploy profiles. Choose Clear Preview when finished, or let normal cleanup restore the device while closing.

Step 5 - Sync Device

Sync validates and saves the local library, reads the device, shows a change summary, asks for confirmation, creates a per-device backup, writes revisioned profile files, commits the index last, and asks the firmware to reload.

Do not unplug during a write
Wait for drive activity to stop. The revisioned write design reduces partial-update risk, but disconnecting USB during filesystem changes is still unsafe.

5. Parent profiles and encoder-turn order

Creating profiles

Command	Result	Worked example
Add	Creates a blank parent named New Profile.	Add one, rename it OBS Studio, then build keys.
Duplicate	Copies every layout, key, label, color, and step.	Duplicate Firefox before creating a testing variant.
Template...	Creates Editing, Media, or Blank from a maintained template.	Choose Media for volume and transport controls.
Import...	Adds one profile from JSON without replacing the library.	Import obs-studio.json from another computer.

Ordering profiles

The left list is the exact encoder-turn sequence. Drag a profile to a new position or use Up and Down. Put profiles used together next to each other. For example, a development cluster might be VS Code, Terminal, Firefox, and ComfyUI.

Clearing, resetting, and deleting

- Clear profile removes assignments but keeps identity and brightness. It asks for confirmation.
- Reset lights enables every key light, restores default idle and pressed colors, and returns brightness to 5 percent.
- Delete removes the selected parent after confirmation. At least one profile must remain.

Profile identity limits

The library supports up to 32 parent profiles. Names are limited to 24 characters. Internal IDs are normalized slugs and remain unique. Use the visible name for human meaning; avoid renaming a parent merely to change a layout screen.

6. Layout screens and encoder-press order

The Screens list in the center shows the primary layout plus up to eight additional layouts. The visible sequence is the order used when pressing the encoder.

Control	Purpose	Example
+	Add a blank additional layout.	Add In App as the fourth Firefox layout.
Duplicate	Copy the selected layout and insert the copy after it.	Duplicate Browse, rename the copy Tabs, then adapt keys.
Up / Down	Move the selected layout earlier or later.	Keep In App last for predictable automatic selection.
Drag	Reorder directly in the Screens list.	Drag Tools before Tabs.
-	Delete an additional layout.	Remove an abandoned Experiment layout. The primary cannot be deleted.

Naming and preview

- Profile name labels the parent chosen by turning the encoder.
- Layout name labels the selected screen shown on the OLED.
- OLED icon adds an optional two-character prefix.
- Light intensity is stored per layout from 0 to 100 percent.

Example: parent Firefox can contain Browse, Tabs, Tools, and In App. The parent name stays Firefox; Layout name changes as you select each screen. Use FF as the icon and 5 percent brightness for each screen unless one needs a deliberate visual warning.

Remembered selection behavior

The device remembers the manually selected layout for each parent profile and restores it after restart. Turning away from Firefox and back does not force Browse again. Automatic In App selection is contextual and does not overwrite the manual selection used by a Manual deck.

7. Key selection, labels, OLED, and RGB

Selecting and moving keys

- Click a key tile to edit that key.
- Ctrl-click multiple key tiles to apply lighting and palette changes together.
- Drag a key to a new position to move it; intervening keys shift to close and open positions.
- Use Swap... to exchange exactly two complete assignments without shifting anything between them.
- Copy control and Paste control duplicate an assignment. Undo and Redo cover editor changes.

Name versus OLED label

Field	Limit and purpose	Example
Name	Up to 24 characters; descriptive text on the GUI tile.	Command palette
OLED label	Up to 6 characters; compact text on the physical display.	CMDPAL
Unset key	Resets name to Unassigned and clears OLED label and steps.	Clear an unused corner key.

Lighting controls

Illuminate this key can disable a key light without losing its saved colors. Idle color is the resting color. Pressed color is short feedback while the key is active. The encoder has no RGB controls.

Recommended color language

Color	Hex	Meaning	Examples
Green	#00FF66	Safe, start, create, confirm, increase	Play, Save, Volume up
Lime	#B8FF00	Navigation and movement	Next tab, Focus right
Yellow	#FFD000	Neutral toggle or reversible edit	Undo, selection
Orange	#FF7A00	Caution or system-changing action	Cut, Volume down
Red	#FF2020	Stop, close, remove, mute, destructive	Interrupt, Close, Shutdown
White	#FFFFFF	Standard pressed feedback	All pressed colors

Palette workflow

1. Select a key whose idle color you want to reuse and choose Save current.
2. Select one or more target keys.
3. Choose the saved hex value in Palette.
4. Choose Selected keys, Current row, or All keys in Apply to.
5. Choose Idle or Pressed.

Example: select Key 4, choose Current row, select #FF2020, and choose Idle. Keys 4-6 receive the red idle color. Use Preview RGB before Sync if brightness or risk semantics matter.

8. Build action sequences

A control may contain up to 16 steps. Steps run top-to-bottom. Add, edit, remove, drag, or use Move up and Move down. Double-clicking a step opens the editor. Safe preview summarizes without sending input; Run on device executes only after confirmation.

The five step types

Type	Fields	Use	Example
hotkey	Up to 6 HID key names	Press keys together.	CONTROL, SHIFT, P
text	Up to 512 characters	Type literal text into the focused application.	pamac checkupdates
consumer	Supported media code	Send an OS media-control event.	VOLUME_INCREMENT
mouse click	LEFT, MIDDLE, or RIGHT button	Click a mouse button.	RIGHT_BUTTON
mouse move	X, Y, wheel from -127 to 127	Move pointer or wheel.	wheel = -1
delay	0 to 10000 ms	Wait between steps.	800 ms after opening a terminal

Hotkeys

Use CircuitPython HID names separated by commas. The searchable key picker appends valid names. CONTROL, C means press both together. If keys must happen sequentially, use separate hotkey steps. Common names include CONTROL, SHIFT, ALT, GUI, ENTER, ESCAPE, TAB, arrows, letters, digits, and F1-F12.

German and US layouts
Hotkey names remain HID names, but typed text and punctuation depend on the Layout selector. Match the configurator to the host desktop layout before synchronization.

Text

Text steps type exactly what is stored. They do not press Enter unless a later hotkey step does so. This distinction is useful for terminal templates: open a terminal, wait, type a command, and stop so the user can review it.

Consumer media actions

Supported codes are MUTE, VOLUME_DECREMENT, VOLUME_INCREMENT, PLAY_PAUSE, SCAN_PREVIOUS_TRACK, SCAN_NEXT_TRACK, STOP, RECORD, and EJECT.

Mouse

Click sends a selected button. Move uses signed X, Y, and wheel values. Small values are easier to control. Test pointer movement in a harmless context because the active application receives it immediately.

Delay

Delays are essential when an earlier step opens or focuses a new window. Start with 800 ms for a terminal or launcher on a typical desktop, then reduce only after repeatable testing. A delay of zero is valid but usually unnecessary.

Presets

The Preset list can fill common actions such as Copy, Paste, Cut, Undo, Redo, Save, Select all, Close tab, Play/Pause, Mute, volume, and mouse clicks. Select Custom action when building a step manually.

9. Worked macro examples

Example A - Copy

Key field	Value
Name	Copy
OLED label	COPY
Idle / pressed	#00FF66 / #FFFFFF
Step 1	hotkey: CONTROL, C

Use Safe preview. It should report one shortcut. Focus a harmless text editor, then Run on device.

Example B - Application launcher

Order	Type	Value	Why
1	hotkey	GUI, D	Open the desktop launcher.
2	delay	300 ms	Allow the launcher to receive focus.
3	text	firefox	Type the application command.
4	hotkey	ENTER	Launch it.

If characters are missing, increase the delay. If the launcher differs on your desktop, replace GUI+D with the configured shortcut.

Example C - Reviewable terminal command

Order	Type	Value
1	hotkey	GUI, ENTER
2	delay	800 ms
3	text	pamac checkupdates

Deliberately no Enter

The MacroPad opens a terminal and types the command but does not execute it. Review, edit, or complete the command manually. Pamac performs its own privilege escalation; do not prefix pamac itself with sudo.

Example D - Volume up

Key field	Value
Name / OLED	Volume up / VOL+
Idle / pressed	#00FF66 / #FFFFFF
Step 1	consumer: VOLUME_INCREMENT

Example E - Right click

Key field	Value
Name / OLED	Context menu / RIGHT
Step 1	mouse click: RIGHT_BUTTON

Example F - Safe delayed shutdown template

Order	Type	Value
1	hotkey	GUI, ENTER
2	delay	800 ms
3	text	shutdown +60

Keep Enter out while testing. Once deliberately enabled, add ENTER as Step 4. Use a red idle color and a clear name such as Shutdown 60m. Add a separate green Cancel key that types shutdown -c and only add Enter after reviewing the complete sequence.

Example G - Multi-step Save As

Order	Type	Value
1	hotkey	CONTROL, SHIFT, S
2	delay	400 ms
3	text	project-copy

Do not add Enter until the target application's dialog behavior is confirmed. Some applications preselect a filename extension or place focus in a different field.

10. Local saving, imports, exports, and backups

Autosave and Save Local

Most edits mark the library dirty and schedule a local save after 750 ms. Save Local writes immediately. Closing with pending local changes asks whether to save, discard, or cancel closing.

One-profile JSON

- Export... writes only the selected parent profile as JSON.
- Import... adds one normalized profile and creates a unique ID if needed.
- Use this format when sharing one profile without palette or unrelated profiles.

Complete library archive

- Export Library writes every profile plus the saved color palette to a .macropad.zip archive.
- Import Library loads the archive into the local editor; review before syncing.
- Use this format for migration, disaster recovery, or sharing a complete setup.

Per-device backups

A changed Sync Device operation creates a backup identified by device UID and timestamp before writing. Backups loads a selected snapshot into the local editor. It does not write the device immediately. Compare the loaded snapshot, then Sync only if restoration is intended.

Goal	Best command	Device changes immediately?
Share one parent profile	Export... / Import...	No

Goal	Best command	Device changes immediately?
Move the complete setup	Export Library / Import Library	No
Return one device to a prior state	Backups, review, then Sync Device	Only at Sync
Capture current device as working copy	Import Device	No

11. Preview, compare, synchronize, and recover

Preview RGB

Preview RGB is temporary and affects only lighting for the selected layout. It is ideal for checking brightness, row grouping, warning colors, and disabled keys. Clear Preview restores normal lighting.

Compare

Compare reads the selected device and reports normalized differences. If comparison fails, verify the device is still mounted, the correct unit is selected, and profile JSON is readable.

What Sync Device does internally

1. Validate and normalize the local project.
2. Save the local library.
3. Read the selected device and compute changes.
4. Stop if there are no changes.
5. Show a change summary and request confirmation.
6. Create a per-device backup.
7. Write revisioned profile files and device_config.json.
8. Commit profiles/index.json and profiles/revision.txt last.
9. Flush filesystem changes and ask firmware to reload.

Interrupted sync recovery

1. Do not repeatedly unplug and reconnect while the drive is active.
2. After the drive settles, reconnect if needed and choose Refresh.
3. Run Device Health.
4. Import Device only if you want the device's surviving state to replace local edits.
5. Otherwise Compare the saved local library and Sync again.
6. If required files are missing, use Repair Firmware after confirming the device.

12. Multiple MacroPads, friendly names, and deck roles

Every connected MacroPad is identified by mount path and UID. Name Device stores a local friendly alias, such as Left Manual Deck or Right App Deck. Import, preview, compare, health, backup, repair, and sync always target the selected device.

Synchronize the same library to two devices

1. Finish and save the local library.
2. Select the first device, Compare, then Sync Device.
3. Select the second device, Compare, then Sync Device.
4. Run Device Health for each if firmware versions might differ.

Manual and App deck roles

Role	Behavior	Example
Manual deck	Stays on the parent and layout selected with its encoder; ignores focused-app commands.	A fixed general-purpose deck for media and system tools.
App deck	Follows the focused desktop application and selects its In App layout.	Firefox controls appear when Firefox gains focus.

On the device, turn to Options or hold the encoder for about one second. Key 1 selects Manual, Key 3 selects App, and encoder press toggles. The role is stored independently on each device and survives restart.

Automatic profile switching

The included user service observes focused i3 or Sway windows and selects matching parents on App decks. Firefox, VS Code, VLC, Discord, LM Studio, ComfyUI, Caja, Krita, LibreOffice, Blender, and other included profiles have matching support.

```
.venv/bin/python tools/install_profile_switcher.py
systemctl --user status macropad-profile-switcher.service
journalctl --user -u macropad-profile-switcher.service -n 50
```

Rules live in `~/.config/macropad-profile-switcher.json`. Restart the user service after editing them. Automatic In App selection does not overwrite the manually remembered layout.

13. Device health, firmware repair, and keyboard layouts

Device Health

Device Health reports configurator version, firmware version, configuration revision, active profile, deck role, keyboard layout, required files, and serial status. Use it before repair because a mounted drive can still have a busy or unavailable serial port.

Repair Firmware

Repair Firmware asks for confirmation, attempts a backup, uses circup to install adafruit_macropad and keyboard-layout dependencies, copies code.py and macropad_core.py, preserves an existing device_config.json and profile library when possible, and reloads the device.

Repair is a device write
Verify the selected friendly name, UID, and mount path. Repair is appropriate for missing required files or a version mismatch, not for a simple key-layout edit.

Keyboard layout

Choose US or German to match the host desktop. The setting affects typed text and characters whose physical HID positions differ. Standard modifier shortcuts are usually familiar, but punctuation and Y/Z behavior can differ. Synchronize after changing layout.

Symptom	Likely cause	Correction
Y and Z are exchanged	Host and configurator layouts differ.	Select the host layout and sync.
Punctuation is wrong	Typed text is mapped through the wrong layout.	Match both configurator and desktop layout.
Ctrl+C works but typed command is wrong	Modifiers are stable while text mapping differs.	Correct Layout, then sync and retest.

14. Limits and validation reference

Item	Limit	Behavior
Parent profiles	32	Additional creation is blocked.
Additional layouts per parent	8, plus the primary	Up to 9 encoder-press screens total.
Profile or layout name	24 characters	Longer input is truncated.
OLED icon	2 characters	Optional prefix.
OLED key label	6 characters	Designed for the physical display grid.
Action steps per control	16	Only the supported maximum is retained and validated.
Hotkey names per step	6	Names are normalized to uppercase HID form.
Text step	512 characters	Longer text is truncated.
Delay	0-10000 ms	Values are clamped.
Mouse X, Y, wheel	-127 to 127	Values are clamped.
Brightness	0-100 percent	Default is 5 percent.
Saved palette	24 colors	Recent unique colors are kept.
Undo history	100 states	Newest editor states are retained.

Validation rejects unnamed profiles, unsupported consumer codes, unsupported click buttons, and invalid normalized projects before synchronization. Invalid colors fall back to defaults. Project and profile imports are normalized before use.

15. Complete control reference

Device toolbar

Control	Description	Example
Device	Select the connected unit targeted by device commands.	Choose Left App Deck before Sync.
Refresh	Scan mounted drives again.	Connect a pad, then Refresh.
Import Device	Replace local editor data with the selected device library.	Capture an existing device before editing.
Preview RGB	Temporarily show selected-layout colors.	Inspect red warning keys.
Clear Preview	Restore normal device lighting.	End a color check without syncing.
Sync Device	Back up and write the complete local library.	Deploy the finished Firefox profile.
Save Local	Save immediately.	Save before closing after many edits.
Layout	Choose host keyboard mapping.	German for a de-DE desktop.
Status	Show autosave and sync state.	Device differs means deployment is pending.

Editing, Library, Device, and File commands

Command	Description	Example
Undo / Redo	Reverse or reapply editor changes.	Undo an accidental drag.
Copy / Paste control	Duplicate a selected assignment.	Copy Ctrl+C into another layout.
Swap...	Exchange exactly two controls.	Swap Key 1 and Key 12.
Compare	List local-to-device differences.	Review before Sync.
Backups	Load a historical device snapshot locally.	Restore yesterday's library after review.
Export / Import Library	Move every profile and palette as an archive.	Migrate to another computer.
Device Health	Inspect versions, required files, and serial.	Diagnose a mounted but unresponsive pad.
Repair Firmware	Reinstall firmware dependencies while preserving profiles when possible.	Restore a missing macropad_core.py.
Name Device	Assign a local alias to a UID.	Left Manual Deck.
Exit	Close with save and preview cleanup.	Ctrl+Q.

Profile and layout controls

Control	Description	Example
Add / Duplicate / Delete	Create, copy, or remove a parent profile.	Duplicate Editing for an alternate map.
Up / Down / drag	Change encoder-turn order.	Put Firefox next to VS Code.
Import... / Export...	Move one profile as JSON.	Share firefox.json.
Template...	Start from Editing, Media, or Blank.	Use Media for transport keys.
Clear profile	Remove assignments but retain identity.	Prepare a duplicate for redesign.
Reset lights	Restore default lighting and 5 percent brightness.	Undo an overbright experiment.
Screens list	Select and order encoder-press layouts.	Browse, Tabs, Tools, In App.
+ / Duplicate / Up / Down / -	Manage additional layout screens.	Duplicate Browse into Tabs.
Profile name / Layout name	Name the parent and selected screen.	Firefox / Tabs.
OLED icon / intensity	Set two-character prefix and per-layout brightness.	FF at 5 percent.

Control editor and macro-step fields

Control	Description	Example
Name / OLED label	Set GUI description and six-character device label.	Command palette / CMDPAL.
Unset key	Clear labels and action steps.	Leave Key 12 unassigned.
Illuminate / Idle / Pressed	Enable light and choose resting and feedback colors.	Green idle, white pressed.
Palette / Apply to / Idle / Pressed	Reuse colors across selected keys, a row, or all keys.	Red idle for a Stop row.
Save current	Store the selected key's idle color.	Save a custom purple.
Step list	Review, select, double-click, and drag execution steps.	Delay before text.
Add / Edit / Remove / Move	Manage the action sequence.	Move terminal delay before text.
Safe preview	Summarize without sending input.	Review shutdown steps.
Run on device	Confirm and execute through the selected pad.	Test Volume Up.
Preset / Type	Choose a common action and its step type.	Copy preset creates CONTROL+C.
Hotkey fields	Enter or search HID names.	CONTROL, SHIFT, S.
Text	Type literal content.	pamac checkupdates.
Consumer	Choose a media code.	PLAY_PAUSE.
Mouse fields	Choose click or signed movement.	RIGHT_BUTTON or wheel -1.
Delay	Pause in milliseconds.	800 ms after opening a terminal.
OK / Cancel	Save or discard the edited step.	Cancel keeps the original delay.

16. Troubleshooting

Problem	Checks	Resolution
No device is listed	CIRCUITPY mounted? Data cable? Correct board ID? Not RPI-RP2?	Reconnect, wait for mount, then Refresh.
Preview or Run fails	Selected device correct? Serial port busy?	Close serial monitors, reconnect, run Device Health.
Import fails	device_config.json and profiles/index.json readable?	Repair malformed files from a known export or use Repair Firmware when required files are missing.
Wrong symbols are typed	Layout selector matches host?	Select US or German correctly, sync, and retest.
Macro types too early	Does a window need focus time?	Insert or increase a delay, often 800 ms.
Macro executes too much	Is ENTER a separate final step?	Remove ENTER and use Safe preview.
New color not on device	Local saved - device differs?	Preview RGB, then Sync Device.
Encoder press editor disabled	Does the parent have multiple layouts?	This is expected; encoder press cycles layouts.
Profile order wrong on device	Was drag order synced?	Compare and Sync Device.
Device stale after sync	Drive activity complete? Firmware reload received?	Wait, reconnect, Refresh, Device Health, Compare.
Firmware files missing	Device Health missing list?	Confirm the selected device, then Repair Firmware.
App deck does not follow focus	Service active? Matching rule? Role set to App?	Inspect systemctl and journalctl output, then restart the service.
Manual choice is overwritten	Device accidentally set to App role?	Open Options and choose Manual deck.

When to use each recovery source

- Use the local library when your latest edits are correct and only the device is stale.
- Use Import Device when the device is authoritative and should replace local edits.
- Use Backups when a previous device snapshot is authoritative.
- Use an exported library archive when migrating or recovering the complete setup.
- Use Repair Firmware for missing runtime files or dependencies, not ordinary profile differences.

17. Operational checklists

Before running a new macro

- Read the step list from top to bottom.
- Use Safe preview.
- Remove ENTER from terminal, shutdown, delete, or reboot templates during initial testing.
- Focus a harmless application or document.
- Confirm the selected device.
- Run on device only after the confirmation text matches your intent.

Before synchronization

- Save Local and confirm the intended parent and layout are selected.
- Select the correct MacroPad by friendly name, UID, and mount path.
- Compare and investigate unexpected removals or large changes.
- Preview RGB when brightness or risk colors changed.
- Keep the USB connection stable until drive activity stops.

Before firmware repair

- Run Device Health first.
- Confirm circup is available in the application environment.
- Confirm the selected device and preserve a complete export when practical.
- Use Repair Firmware only for firmware/dependency problems.

Glossary

Term	Meaning
Parent profile	A top-level profile selected by turning the encoder.
Layout / subprofile	One 12-key screen within a parent, selected by encoder press.
Local library	The GUI working copy stored in the computer's application-data folder.
Device library	The profile set currently stored on the selected CIRCUITPY drive.
Control	One of 12 keys or the single-layout encoder-press assignment.
Step	One hotkey, text, consumer, mouse, or delay operation in a control sequence.
HID name	CircuitPython key identifier such as CONTROL, ENTER, or PAGE_DOWN.
Consumer code	Operating-system media command such as MUTE or PLAY_PAUSE.
Preview RGB	Temporary lighting display that does not deploy profile data.
Sync	Validated, backed-up write of the local library to the selected device.
Manual deck	Device role that stays on encoder-selected profiles.
App deck	Device role that follows focused-application rules.
In App	Contextual layout selected by the focus service without replacing manual memory.

18. Final quick-start recipe

1. Connect the MacroPad and choose Refresh.
2. Select the correct device.
3. Import Device if it contains profiles to preserve.
4. Create or select a parent profile and layout.
5. Select a key; set Name, OLED label, idle color, and pressed color.
6. Add macro steps and use Safe preview.
7. Save Local.
8. Compare with the device.
9. Preview RGB if lighting changed.
10. Sync Device and wait for drive activity to finish.
11. Test the physical key in a harmless context.

You now have a recoverable workflow

Local autosave protects editor work, Compare exposes differences, Preview RGB checks lighting without deployment, Sync creates a per-device backup, and complete Library Export provides a portable recovery point.